Serie numeriche

Def.

Sia $\{a_n\}$ una successione di numeri reali, si dice serie la successione dei suoi termini (sono infiniti).

$$\sum_{k=0}^{\infty} a_k$$

Chiamamo somme parziali $\{s_n\}$ la serie dei termini della successione fino a $n \in \mathbb{N}$.

$$S_n = \sum_{k=0}^{n} a_k = a_0 + a_1 + a_2 + \cdots + a_{n-1} + a_n$$

$$S_0 = a_0$$

$$S_n = S_{n-1} + a_n = S_{n+1} - a_{n+1}$$

Per capire come si "comporta" una serie, prendiamo $n \to \infty$

$$\sum_{k=0}^{\infty} a_k = \lim_{n \to +\infty} S_n = \lim_{n \to +\infty} \sum_{k=0}^{n} a_k$$

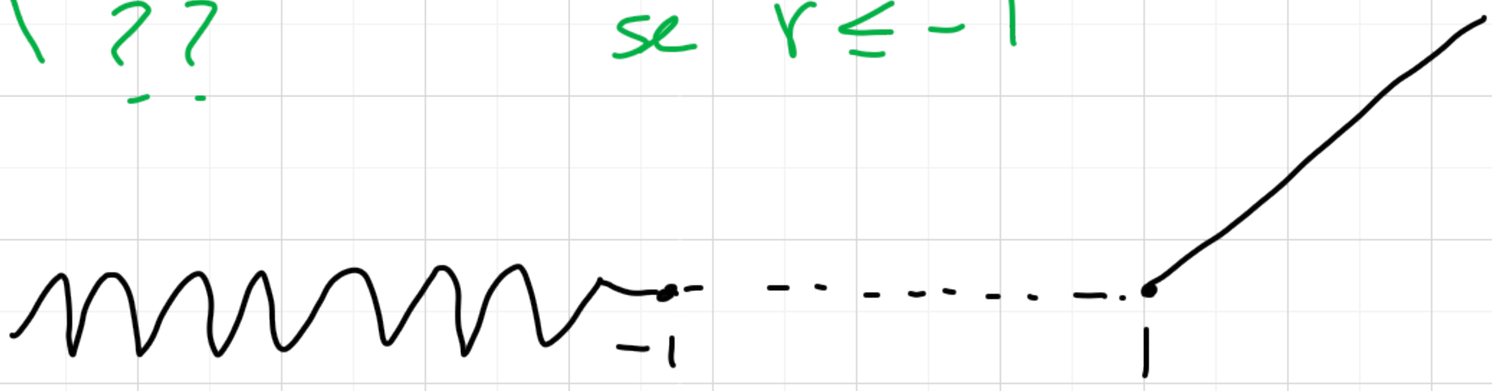
- CONVERGE
- DIVERGE
- $\nexists$ (IRREGOLARE)

Serie geometrica

$$\sum_{k=0}^{\infty} r^k$$

$a_k = r^k$

- $\frac{1}{1-r}$ se $-1 < r < 1$
- $+\infty$ se $r \geq 1$
- ?? se $r \leq -1$



+ GENERALE

$$\sum_{k=k_0}^{\infty} ar^k = \frac{ar^{k_0}}{1-r} \quad \text{se} \quad r \in (-1, 1)$$

Convergenza

Esempio

$$\sum_{k=2}^{\infty} \frac{5}{2^k} = \frac{5 \cdot \left(\frac{1}{2}\right)^2}{1 - \frac{1}{2}} =$$

$k_0 = 2$
$a = 5$
$r = \frac{1}{2} \in (-1, 1)$

i)

$$= \frac{5}{\cancel{4}_2} \cdot \cancel{2} = \frac{5}{2}$$

$r = \frac{1}{2}$

ii)

$$\sum_{k=2}^{\infty} \frac{5}{2^k} = 5 \cdot \sum_{k=0}^{\infty} \frac{1^k}{2^k} - \frac{5}{2^0} - \frac{5}{2^1} =$$

(a_0, a_1)

$$= 5 \cdot \frac{1}{1 - \frac{1}{2}} - 5 - \frac{5}{2} = 5 \cdot 2 - 5 - \frac{5}{2} = \frac{10 - 5}{2}$$

Esempio

$$\sum_{k=1}^{\infty} \frac{11^k}{10^{k+2}} = \frac{1}{10^2} \cdot \sum_{k=1}^{\infty} \left(\frac{11}{10}\right)^k$$

$r = \frac{11}{10} > 1$

$\Rightarrow$ DIVERGE a $+\infty$

Esempio

$$\sum_{k=0}^{\infty} \frac{8^k}{\pi^{2k}} = \sum_{k=0}^{\infty} \left(\frac{8}{\pi^2}\right)^k =$$

$r = \frac{8}{\pi^2} = 0{,}81$

$$= \frac{1}{1 - \frac{8}{\pi^2}} = \frac{1}{\frac{\pi^2 - 8}{\pi^2}} = \frac{\pi^2}{\pi^2 - 8} = \frac{5}{2}$$

Sviluppi decimali

$$0,\overline{3} = 0,33333\ldots = 0,3 + 0,03 + 0,003 + \cdots$$

$$\frac{3}{10} \quad \frac{3}{10^2} \quad \frac{3}{10^3} \quad \ldots$$

$$= 3 \cdot \sum_{k=1}^{\infty} \frac{1}{10^k} = 3 \cdot \left[\sum_{k=0}^{\infty} \frac{1}{10^k} - \frac{1}{10^0} \right]$$

$$= 3 \left[\frac{1}{1 - \frac{1}{10}} - 1 \right] = 3 \left[\frac{1}{\frac{9}{10}} - 1 \right] = 3 \cdot \left[\frac{1}{}

$$\left[\frac{3^{10}}{3^{9}} - 1\right] = 3 \cdot \frac{1}{3^{9}} = \frac{1}{3}$$

$$3,8142\overline{5} = 3,814 + 0,00025\dot{2}5\dot{2}5 =$$

$$= \frac{3814}{1000} + \frac{25}{10^5} + \frac{25}{10^7} + \frac{25}{10^9} \cdot$$

$$= \frac{3814}{1000} + \sum_{k=2}^{\infty} \frac{25}{10^{2k+1}}$$

$$= \frac{3814}{1000} + \frac{25}{10} \sum_{k=0}^{\infty} \left(\frac{1}{10^2}\right)^k - \frac{25}{10} - \frac{2}{1}$$

$$= \frac{381425 - 3814}{99000} = \frac{377611}{99000}$$

... =

$$3,81425 = \frac{z\,xxx\,yy - z\,xxx}{(9\cdot y)(0\cdot x)}$$

z xxx yy

intero antiperiodo periodo

$$\frac{5}{000} =$$



$$0,\overline{9} = 1$$



NON VALE SE ULTIMA CIFRA = 9

Serie armonica generalizzata

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$$

- CONVERGE se $\alpha > 1$
- DIVERGE $+\infty$ se $\alpha \leq 1$

Serie telescopiche

$$\sum_{k=0}^{\infty} \left(b_k - b_{k+1} \right) = b_0 - \lim_{k \to \infty} b_k$$

$$\sum_{k=0}^{\infty} \left(b_k - b_{k+m} \right) = \sum_{k=0}^{m} b_k - m \lim_{k \to \infty} b_k$$

SERIE MENGOLI

LOGARITMI CON FRAZIONI

SIA K e K+m

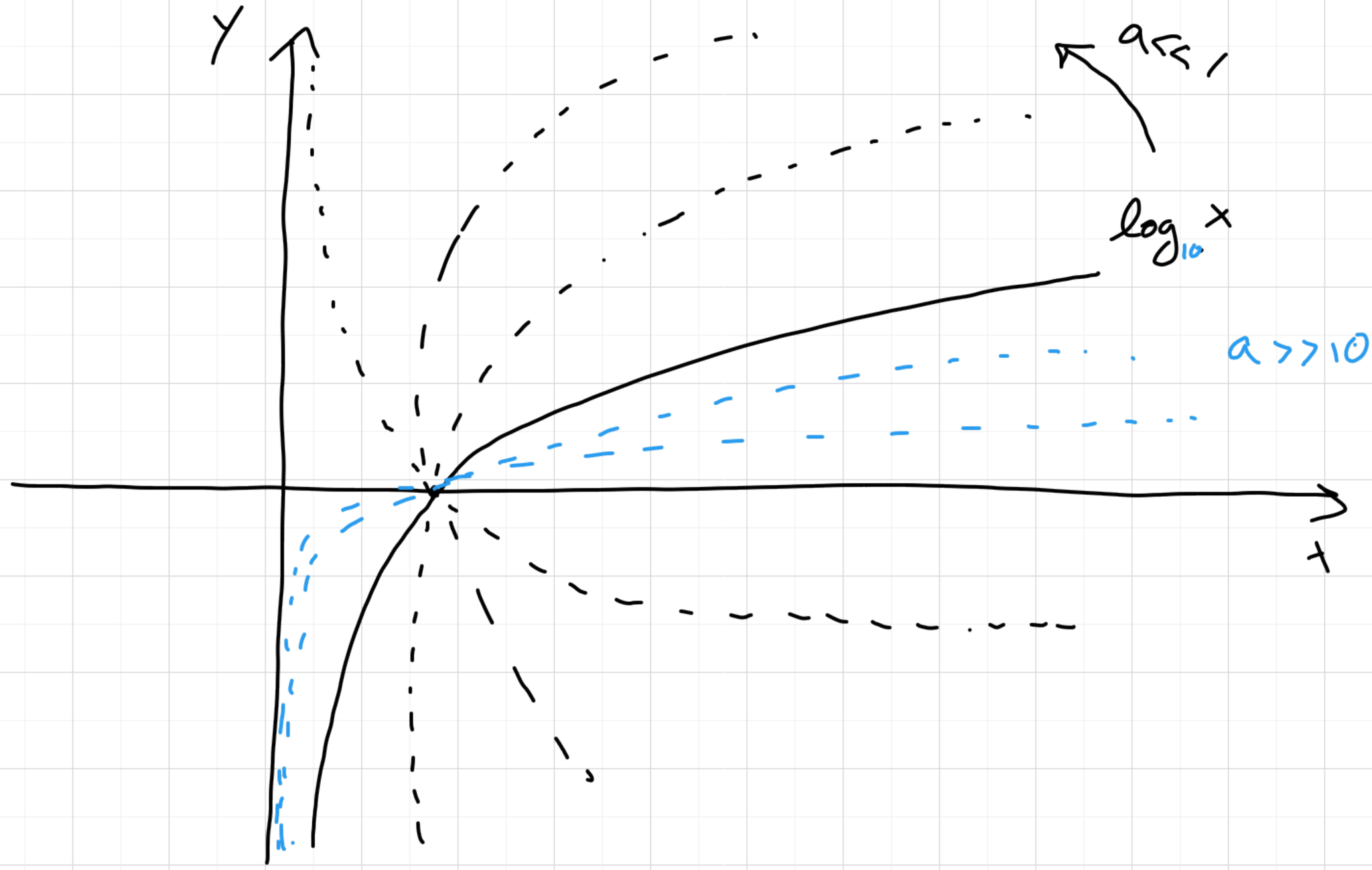
Ese) $\sum_{k=1}^{\infty} \log_{\frac{1}{2}}\left(\frac{k}{k+3}\right) =$

$b_k = \log_{\frac{1}{2}} k$
$m = 3$

$= \sum_{k=1}^{\infty} \left(\log_{\frac{1}{2}}(k) - \log_{\frac{1}{2}}(k+3) \right) =$

$= \sum_{k=1}^{\infty} \left(\log_{\frac{1}{2}}(k) - \log_{\frac{1}{2}}(k+3) \right)$

$= \underbrace{\log_{\frac{1}{2}} 3}_{b_3} + \underbrace{\log_{\frac{1}{2}} 1}_{b_1} + \underbrace{\log_{\frac{1}{2}} 2}_{b_2} - 3 \overbrace{\left(\lim_{k\to\infty} \log_{\frac{1}{2}} k \right)}^{m \cdot \lim_{k\to\infty} b_k} = -3(-$



$\infty) = +\infty$



$$\sum_{k=1}^{\infty} \frac{1}{k(k+2)}$$

Serie con termini non negativi

A. Criterio confronto
- diretto: $0 \le a_k \le b_k$
- asintotico: $a_k \sim b_k$

B. Criterio radice: $\lim_{k \to +\infty} \sqrt[k]{a_k}$

C. Criterio rapporto: $\lim_{k \to +\infty} \frac{a_{k+1}}{a_k}$

$\Big\} = l$

Convergencia serie

Se $\sum_{k=0}^{\infty} a_k$ converge $\Rightarrow \lim_{k\to\infty} a_k = 0$

$\ell < 1$ CONVERGE

$\ell > 1$ DIVERGE

$\ell = 1$??

Es. $\sum_{k=1}^{\infty} \frac{2^k k!}{k^k}$

✓ TERMINI POSITIVI

⚠ FATTORIALE

$\lim_{k\to\infty} \frac{a_{k+1}}{a_k} = \lim_{k\to\infty} \frac{2^{k+1}(k+1)!}{(k+1)^{k+1}} \cdot \frac{k^k}{2^k k!} =$

$= \lim_{k\to\infty} \frac{2^k \cdot 2^1 \cdot (k+1)\,k!}{(k+1)^k (k+1)} \cdot \frac{k^k}{2^k k!} =$

$= 2 \lim_{k\to\infty} \frac{k^k}{(k+1)^k} = 2 \lim_{k\to\infty} \left(\frac{k}{k+1}\right)^k = 2 \cdot \lim_{k\to\infty} \left[\frac{k}{k}\right.$

$$= 2 \lim_{k\to\infty} \frac{1}{\left(1+\frac{1}{k}\right)^k} = 2 \cdot \frac{1}{e} = \frac{2}{e} = \ell < 1$$

$\downarrow$

$b_k \to 0$ per $k \to \infty$

$$\Longrightarrow \sum_{k=0}^{\infty} a_k \quad \text{CONVERGE}$$

$$\left[\frac{k}{\left(1+\frac{1}{k}\right)}\right]^k =$$

Es. $\sum_{k=1}^{\infty} \left(1 - \frac{1}{k}\right)^{k^2}$

✓ TERMINI POS.

⚠ POTENZA

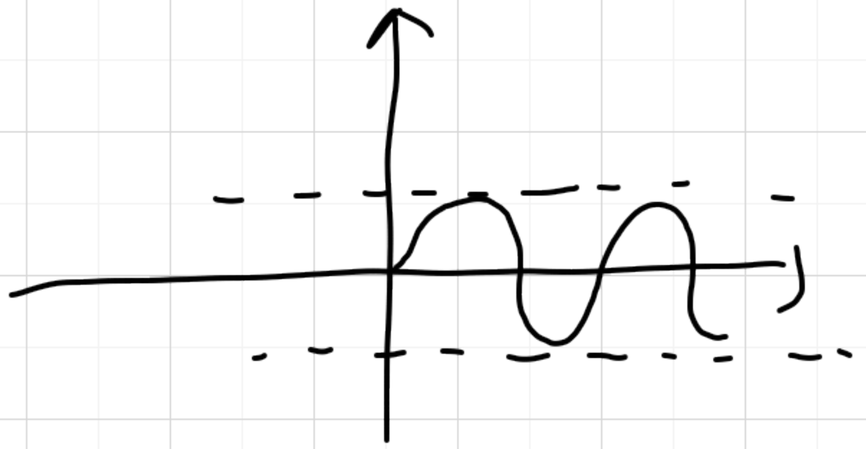
$$\lim_{k\to\infty} \sqrt[k]{a_k} = \lim_{k\to\infty} \sqrt[k]{\left(1 - \frac{1}{k}\right)^{k^2}} =$$

$$= \lim_{k\to\infty} \left(1 - \frac{1}{k}\right)^{\frac{k^2}{k}} = \lim_{k\to\infty} \left(1 + \frac{(-1)}{k}\right)^{k} = 0$$

$$= e^{-1} = \frac{1}{e} < 1 \Rightarrow \text{CONVERGE}$$

Esempio

$$\sum_{k=1}^{\infty} \frac{|\sin k|}{k^2}$$



$$|\sin k| \leq 1$$

$$\leq \sum_{k=1}^{\infty} \frac{|\sin k|}{k^2} \leq \sum_{k=1}^{\infty} \frac{1}{k^2}$$

a_k b_k

(serie armonica con $\alpha = 2 > 1$ ↓ CONVERGE)

Esempio

$$\sum_{k=1}^{\infty} \log\left(1+\frac{1}{k^2}\right) \sim \frac{1}{k^2}$$

$\hookrightarrow$ serie armonica $\alpha = 2 > 1$

$\Rightarrow$ CONVERGE

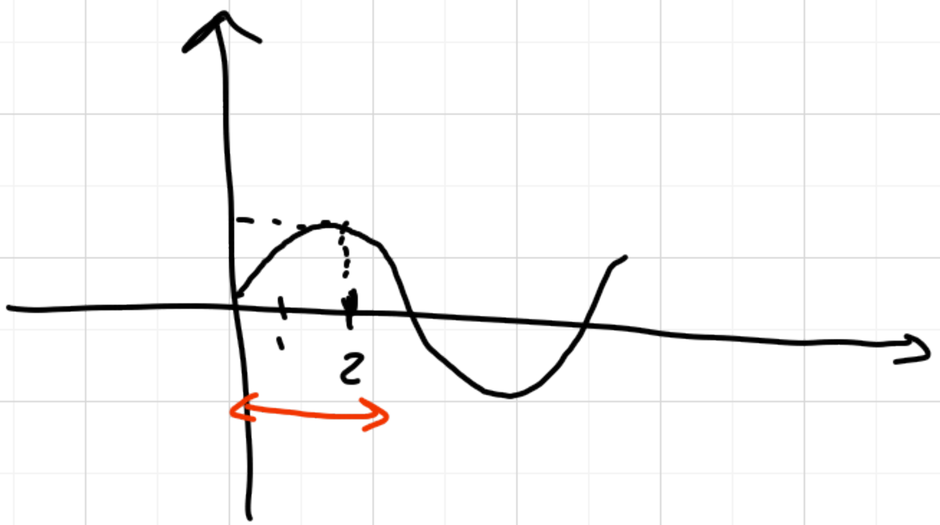
Esempio

$$\sum_{k=1}^{\infty} \sin\left(\frac{2}{k}\right) \sim \frac{2}{k}$$

$\to 0$ per $k \to \infty$

serie armonica $\alpha = 1$

$\Rightarrow$ DIVERGE



Esercizio

$$\sum_{k=1}^{\infty} \frac{2+\sin k}{k^2 \log k}$$

a_k

$$\frac{2+\sin k}{k^2 \log k} \leq \frac{2+1}{k^2 \cdot k} = \frac{3}{k^3}$$

b_k

$\sin k \leq 1$

$\log k \leq k$

↳ serie armonica

$\alpha = 3$

$\Rightarrow$ CONVERGE b_k

$\Rightarrow$ CONVERGE a_k

Esempio $\sum_{k=2}^{\infty} (\log k)\left(\log\left(1+\frac{1}{k}\right)\right) \sim \log k \cdot \frac{1}{k}$

$\lim_{k\to\infty} \frac{\log k}{k} = 0$

Converge $\Rightarrow \lim_{k\to+\infty} a_k = 0$

$\lim_{k\to\infty} a_k = 0 \not\Rightarrow$ CONVERGE

$$\frac{\log^{k}}{k} \geq \frac{1}{k}$$

b_k $\qquad$ a_k

↳ DIVERGE

$\Rightarrow$ b_k DIVERGE

Esempio

$$\sum_{k=1}^{\infty} \frac{k}{\log^2 k + 2}$$

$$\lim_{k \to \infty} a_k \not\to 0 \implies \text{DIVERGE}$$

Esempio $\sum_{k=1}^{\infty} \frac{k\left(\log\left(e^4+\frac{1}{k}\right)\right)^k}{5^k+3}$

$$\lim_{k\to\infty} \sqrt[k]{a_k} = \lim_{k\to\infty} \sqrt[k]{\frac{k\left[\log\left(e^4+\frac{1}{k}\right)\right]^k}{5^k+3}} =$$

$$= \lim_{k\to\infty} \log\left(e^4+\frac{1}{k}\right) \sqrt[k]{\frac{k}{5^k\left(1+\frac{3}{5^k}\right)}} =$$

$$\lim_{k\to+\infty} \frac{\log\left(e^4+\frac{1}{k}\right)}{5} \cdot \sqrt[k]{\frac{k}{1+\frac{3}{5^k}}} =$$

$$\lim_{k\to\infty} \frac{\log\left(e^4-\frac{1}{k}\right)}{5} \cdot \frac{\sqrt[k]{k}\ ^{*}}{\sqrt[k]{1+\frac{3}{5^k}}} = \frac{4\cdot 1}{5\cdot 1} =$$

$$^{*}\ \sqrt[k]{k} = k^{\frac{1}{k}} = e^{\log k^{\frac{1}{k}}} = e^{\frac{1}{k}\log k} =$$

$$= e^{\frac{\log k}{k}} \longrightarrow e^0 = 1 \quad \text{per } k\to\infty$$

∞^0

Esc

$$\sum_{k=1}^{\infty} \left[\left(1 + \frac{5}{k} \right)^{k} - 1 \right]$$

$$\lim_{k \to \infty} a_k = e^5 - 1 \neq 0$$

$\Longrightarrow$ DIVERGE

$= \frac{4}{5} < 1$

$\Rightarrow$ CONVERGE



$$\sum_{K=2}^{\infty} \frac{2^{K+1} x^{K-1}}{(K-1)!} = -3$$

Trova il valore di x

$$* \quad \sum_{k=0}^{\infty} \frac{a^k}{k!} = e^a \qquad \sum_{k=0}^{\infty} \frac{1}{k!} = e$$